

Paper	ENCN342 – Fluid Mechanics and Hydraulics
Topic	Steady Open Channel Flow
Module	2 – Specific Energy and Critical Flow

Summary

This module uses the energy and continuity equations to analyse steady flows in open channels where energy losses are negligible. In particular the transition problem is analysed and the concept of critical flow introduced.

Prerequisites

Module 1.

Objectives

At the conclusion of this module you should:

1. Understand the concept of specific energy and be able to plot the relationship between specific energy and flow depth for varying discharges.
2. Be able to analyse the transition problem using the specific energy diagram, and understand the issues surrounding flow regime accessibility.
3. Understand the special properties of critical flow and be able to deduce the relationship between discharge and flow depth at critical conditions.
4. Be able to work with the specific energy diagram in dimensionless form to solve flow transition problems.
5. Understand the concept of a flow control, and realise that critical flow is one example of such a control.
6. Understand the concept of choked flow and its implications.

References

Hunt 1995 pp 12.1-12.12

Key Points

1. Specific energy

The **specific energy** is defined to be

$$E = y + \frac{u^2}{2g} \quad (1)$$

or

$$E = y + \frac{q^2}{2gy^2} \quad (2)$$

where q is the discharge per unit width. The total energy is then the sum of the bed elevation and the specific energy.

$$E + z = H = \text{constant} \quad (3)$$

The E - y curve, given by equation (2), can be used to analyse steady open channel flow where energy losses are negligible. The key characteristics of the relationship given by equation (2) are

- for particular values of E and q there can be 0, 1 or 2 solutions for y . When there are two solutions these are called **alternate depths**,
- one solution corresponds to a shallow, fast flow and the second to a deep, slow flow.

2. Flow regime accessibility

If a channel flow undergoes a transition from one flow regime to another, the new flow regime must be attainable from the original by traversing a physically acceptable path on the E - y diagram. For example, if the flow corresponds to a constant discharge per unit width, then the path taken by the flow through the transition must lie on a curve of constant q . A jump from one branch of this curve to another is not generally possible. It may only occur if the assumptions leading to the energy equation are invalid in part of the flow, eg the strong vertical accelerations under a sluice gate. Each flow problem should be carefully analysed through the use of the E - y diagram.

3. Critical flow

The minimum point on the E - y diagram corresponds to **critical flow** conditions. At critical flow the specific energy is a minimum for a particular discharge (or alternatively the discharge is maximum for a particular specific energy). The conditions for critical flow are

$$y_c = \sqrt[3]{\frac{q^2}{g}} \quad (4)$$

or

$$u_c^2 = gy_c \quad (5)$$

The relationship between the critical flow depth and the specific energy is

$$y_c = \frac{2}{3} E_c \quad (6)$$

Critical flow corresponds to flow conditions in which the **Froude number** based on the flow depth is 1.

$$\sqrt{\frac{u_c^2}{gy_c}} = Fr_c = 1 \quad (7)$$

and, as the speed of a long surface wave is given by

$$c = \sqrt{gy} \quad (8)$$

the Froude number can be thought of as the ratio of the flow speed to the wave speed.

The deep, slow flows referred to above are called **subcritical** flows, and the Froude number for such flows is less than 1. The fast, shallow flows are termed **supercritical**, and correspond to a Froude number greater than 1.

4. E-y diagram in dimensionless form

If equation (2) is non-dimensionalised by dividing by the critical flow depth, y_c , then the relationship between the non-dimensional specific energy E' and non-dimensional flow depth y' , defined in equations (9)

$$E' = \frac{E}{y_c} \quad \text{and} \quad y' = \frac{y}{y_c} \quad (9)$$

is

$$E' = y' + \frac{1}{2y'^2} \quad (10)$$

Two iterative schemes can be used to obtain the non-dimensional flow depth when the non-dimensional specific energy is known.

For subcritical flow

$$y'_{n+1} = E' - \frac{1}{2y_n'^2} \quad (11)$$

For supercritical flow

$$y'_{n+1} = \sqrt{\frac{1}{2(E - y'_n)}} \quad \text{Note that it must read } E' \text{ in equation (12)} \quad (12)$$

The relationship in equation (10) is plotted in Fig 2.1.

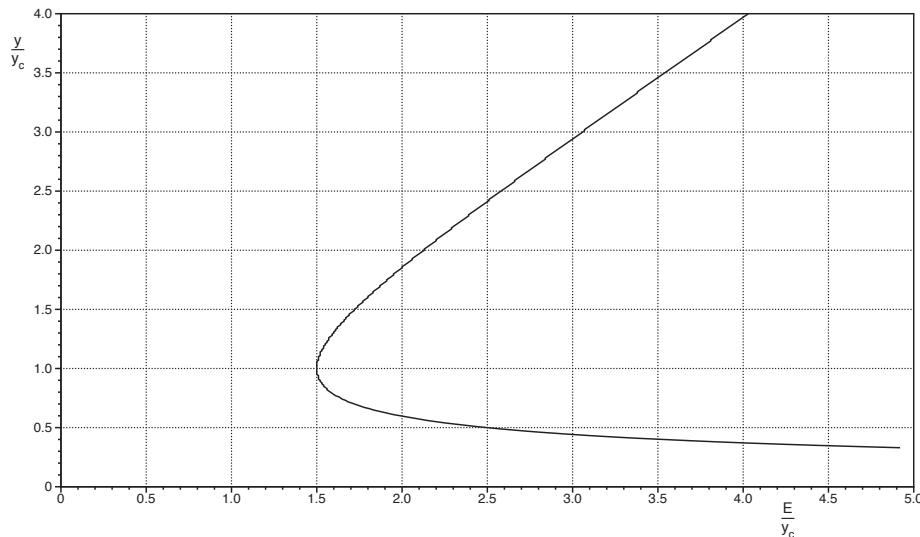


Figure 2.1 The dimensionless E' - y' curve

5 Flow Controls

For a particular discharge there is no unique solution for the flow depth and specific energy unless some type of **control** is “placed” in a channel. This control can take many forms

- friction, or
- a localised feature such as a sluice gate, or
- a critical flow section

A control determines a **unique depth-discharge relationship**. For a sluice gate this relationship is

$$q = C_c H_g y_1 \sqrt{\frac{2g}{y_1 + C_c H_g}} \quad (13)$$

where y_1 is the upstream flow depth, C_c is the contraction coefficient (typically 0.6) and H_g is the height of the sluice gate above the channel bed.

6 Choke

A **choke** occurs in a flow when a channel transition is so extreme that no flow regime is energetically possible downstream. In this case the specific energy is below the minimum value given by critical conditions, for the channel’s discharge. A choke will

modify the upstream flow until a physically acceptable flow regime downstream is possible. Generally this involves a surge wave travelling upstream, increasing the upstream flow depth, until the specific energy is sufficient to allow a steady flow to be re-established. The flow through the choke will now be critical.

Worked Examples

Question 1

Water is flowing at a velocity of 3m/s and a depth of 3m in a channel of rectangular cross-section. Find the change in depth, and in absolute water level produced by

- (a) a smooth upward step of 0.3m in the channel bed
- (b) a smooth downward step of 0.3m in the channel bed.
- (c) Also find the maximum allowable size of upward step for the upstream flow to be possible as specified (ie so it does not choke the flow).

Solution

The upstream conditions for this flow are as follows:

$$y = 3\text{m}$$

$$u = 3\text{m/s}$$

$$q = 9\text{m}^2/\text{s}$$

$$E = y + \frac{q^2}{2gy^2} = 3 + \frac{81}{176.5} = 3.46\text{m}$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 2.02\text{m}$$

$$E' = 1.71 \quad \text{and} \quad y' = 1.49$$

- (a) An upward step in bed elevation will cause a decrease in specific energy by the same amount. Therefore downstream the specific energy is

$$E = 3.16\text{m} \quad \text{and} \quad E' = 1.56$$

As the lowest possible value of E' is 1.5 at critical flow, the new flow regime is only slightly above critical.

To find the new flow depth we can use the iterative scheme for subcritical flow. Start the iteration with the upstream flow depth.

n	y'_n
0	1.49
1	1.33
2	1.28
3	1.25
4	1.24
5	1.23
6	1.23

Thus the dimensionless flow depth is 1.23, which corresponds to a flow depth of 2.48m.

Thus the flow depth has dropped 0.52m and the absolute water level has dropped 0.52-0.3 = 0.22m. So we have this apparent anomaly that the flow depth drops as the flow passes over an upward step.

(b) If the channel bed level drops then the specific energy must increase by the same amount. Therefore downstream the specific energy is

$$E = 3.76m \text{ and } E' = 1.86$$

To find the new flow depth we can use the iterative scheme for subcritical flow. Start the iteration with the upstream flow depth.

n	y'_n
0	1.49
1	1.63
2	1.67
3	1.68
4	1.68

Thus the dimensionless flow depth is 1.68, which corresponds to a flow depth of 3.40m.

Thus the flow depth has increased 0.4m and the absolute water level has increased 0.4-0.3 = 0.1m. Here we have the opposite of the previous example. The flow depth increases as the flow experiences a downward step in bed elevation.

(c) Finally we are to calculate the maximum upward step that does not affect the upstream flow. This step will correspond to a critical flow downstream. This is the minimum specific energy for the given discharge that will sustain the flow. If the step is greater than this the flow will be **choked** as there is insufficient energy for the flow to pass over the step.

We know that at critical flow the specific energy is 1.5 times the depth, or $E' = 1.5$. Thus for this flow, which has a critical depth of 2.02m, the specific energy at critical is 3.03m. The upstream specific energy is 3.46m, so the maximum upward step that will not affect the upstream flow is

$$\Delta z = E_{upstream} - E_c = 3.46 - 3.03 = 0.43m$$

Question 2

Water is flowing at a velocity of 5m/s and a depth of 0.3m in a channel of rectangular cross-section that has a width of 3m. Find the change in depth, and in absolute water level produced by

(a) a smooth contraction in channel width of 0.3m

- (b) a smooth expansion in channel width of $0.3m$.
- (c) Also find the greatest allowable contraction in width for the upstream flow to be possible as specified.

Solution

The upstream conditions for this flow are as follows:

$$y = 0.3m$$

$$u = 5m/s$$

$$q = 1.5m^2/s$$

$$Q = qB = 1.5 \times 3 = 4.5m^3/s$$

$$E = y + \frac{q^2}{2gy^2} = 0.3 + \frac{1.5^2}{2 \times 9.81 \times 0.3^2} = 1.57m$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 0.61m$$

$$E' = 2.57 \quad \text{and} \quad y' = 0.49$$

- (a) A contraction in channel width does not affect the value of specific energy but it does affect the discharge per unit width, and therefore it changes the non-dimensional specific energy. After the contraction we have the following conditions

$$q = \frac{Q}{B} = \frac{4.5}{3-0.3} = 1.67m^2/s$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 0.66m$$

$$E' = \frac{E}{y_c} = \frac{1.57}{0.66} = 2.38$$

To find the new flow depth we can use the iterative scheme for supercritical flow (clearly $Fr > 1$ here). Start the iteration with the upstream flow depth.

n	y'_n
0	0.49
1	0.51
2	0.52
3	0.52

Thus the dimensionless flow depth is 0.52, which corresponds to a flow depth of $0.34m$.

Thus the flow depth has increased $0.04m$ and the absolute water level has increased the same amount.

(b) If the channel expands by $0.3m$ then the discharge per unit width will drop. Downstream the conditions are now

$$q = \frac{Q}{B} = \frac{4.5}{3+0.3} = 1.36m^2 / s$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 0.57m$$

$$E' = \frac{E}{y_c} = \frac{1.57}{0.57} = 2.75$$

To find the new flow depth we again use the iterative scheme for supercritical flow. Start the iteration with the upstream flow depth.

n	y'_n
0	0.49
1	0.47
2	0.47

Thus the dimensionless flow depth is 0.47, which corresponds to a flow depth of $0.27m$.

Thus the flow depth has decreased $0.03m$ and the absolute water level has decreased by the same amount.

(c) Finally we are to calculate the maximum contraction in channel width that will not affect the upstream flow. Thus we need to calculate the contraction that will cause the flow in the contraction to be critical. Any further reduction in channel width will choke the flow, and generate a surge wave upstream.

We know that at critical flow the specific energy is 1.5 times the depth, or $E' = 1.5$. As the specific energy is constant through the contraction E' can be reduced through an increase in critical depth. This increase in critical depth will be due to an increase in q as the channel width is contracted.

So for critical flow we have

$$E' = \frac{E}{y_c} = \frac{1.57}{y_c} = 1.5$$

so

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \frac{1.57}{1.5} = 1.05m$$

Therefore within the contraction q is given by

$$q = \sqrt{1.05^3 g} = 3.37 = \frac{Q}{B} = \frac{4.5}{B}$$

and this allows us to calculate the flow width that will create critical flow.

$$B = \frac{4.5}{3.37} = 1.34m$$

Therefore the maximum contraction allowed is 1.66m.

Question 3

A sluice gate is placed across a canal that is 5m wide and carries a discharge of 3cumecs. The sluice gate is lowered until the downstream flow depth is 0.2m. Calculate the depth upstream of the gate.

Solution

Energy losses through the gate are negligible so the specific energy is the same at both the upstream and downstream sections. However, it is important to realise that near the gate itself the energy equation is invalid due to the high vertical accelerations and inappropriate assumption of hydrostatic pressure in the vertical.

We know the flow upstream of the gate is subcritical and the downstream flow is supercritical. Somewhere near the gate the flow is critical and therefore the gate is a control.

Through the gate the discharge per unit width does not change and is given by

$$q = \frac{Q}{B} = \frac{3}{5} = 0.6m^2 / s$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 0.33m$$

The specific energy of the flow can be calculated from the downstream conditions

$$E = y + \frac{q^2}{2gy^2} = 0.2 + \frac{0.6^2}{2 \times 9.81 \times 0.2^2} = 0.66m$$

and the non-dimensional specific energy is

$$E' = \frac{E}{y_c} = \frac{0.66}{0.33} = 2.00$$

To find the upstream conditions we need to calculate the alternate depth to 0.2m for the same specific energy, or the same non-dimensional specific energy. The alternate depth will be for subcritical flow and therefore the iterative scheme we need to use should be for subcritical flow, and we will need to take a guess at the starting value for the process (we can't use the downstream depth as it is supercritical). We will

choose $y' = 2$ as an estimate for the upstream depth (this is estimated from the $E'-y'$ diagram).

n	y'_n
0	2.00
1	1.88
2	1.86
3	1.86

Thus the upstream non-dimensional depth is 1.86, and the dimensional depth is $0.61m$.

Question 4

Water flows out of a lake into a steep rectangular channel that is $4m$ wide. The lake level above the channel bed at the outfall is $3m$. Find the discharge from the lake.

Solution

The flow in the lake is subcritical, but downstream of the outfall, due to the steepness of the channel, the flow will be supercritical. Therefore at the outflow section, the flow must be critical. We saw in the notes that a transition from subcritical flow to supercritical flow can be made only if the channel bed elevation changes or the flow is critical. Thus, theoretically, at the point where the channel bed has zero slope the flow must be critical, and a measurement of the depth at this point will provide the flow velocity or discharge per unit width (remember that a critical section is a control in the sense that a depth-discharge relationship is specified at that section).

So for this problem

$$q = \sqrt{gy^3} = \sqrt{9.81 \times 3^3} = 16.3m^2 / s$$

and the total discharge is

$$Q = qB = 16.3m^2 / s \times 4m = 65.2m^3 / s$$

Question 5

Road works along a stretch of river require the riverbank to be extended into the river over a short section of the river in order to allow for the road expansion. The river is $4m$ wide at the section where the contraction in river width is planned, and the average river flow is $7cumecks$, at a depth of $1.2m$.

- Determine the maximum contraction in channel width that can be undertaken without affecting the upstream flow.
- If peak flow for the river is $20cumecks$ at a depth of $2.7m$, will this flow be affected upstream by the changed river conditions calculated in (a) (assuming no overflow of the banks).

Solution

The upstream flow conditions are

$$y = 1.2m$$

$$Q = 7m^3 / s$$

$$q = \frac{Q}{B} = \frac{7}{4} = 1.75m^2 / s$$

$$E = y + \frac{q^2}{2gy^2} = 1.2 + \frac{1.75^2}{2 \times 9.81 \times 1.2^2} = 1.31m$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 0.68m$$

$$E' = 1.92 \quad \text{and} \quad y' = 1.76$$

(a) If the channel's width is reduced the specific energy will remain the same (provided the transition is smooth), but the discharge per unit width will increase. We wish to know how much q can increase before the upstream conditions are affected. The maximum constriction corresponds to critical flow, so under these conditions the specific energy is 1.5 times the critical depth. Thus

$$y_c = \frac{2}{3}E = 0.87m$$

This corresponds to a discharge of

$$q = \sqrt{gy_c^3} = 2.54m^2 / s$$

Given the total discharge of *7cumecs* this value of q corresponds to a flow width of *2.76m*. Thus the channel bank can be extended *1.24m* into the river without affecting the upstream flow conditions.

(b) Now we are concerned about whether the channel contraction in (a) will have an adverse affect on peak flow conditions (assuming the river doesn't over flow its banks through the contraction. In peak flow conditions the upstream parameters are

$$y = 2.7m$$

$$Q = 20m^3 / s$$

$$q = \frac{Q}{B} = \frac{20}{4} = 5m^2 / s$$

$$E = y + \frac{q^2}{2gy^2} = 2.7 + \frac{5^2}{2 \times 9.81 \times 2.7^2} = 2.87m$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 1.37m$$

$$E' = 2.10 \quad \text{and} \quad y' = 1.97$$

As the flow is further from critical at peak conditions we expect that the constriction will have less influence. We can check this by calculating the value of E' downstream of the constriction calculated in (a).

$$q = \frac{Q}{B} = \frac{20}{2.76} = 7.25m^2/s$$

so

$$y_c = \sqrt[3]{\frac{q^2}{g}} = 1.75m$$

and

$$E' = \frac{E}{y_c} = \frac{2.87}{1.75} = 1.64$$

which is above the critical value of 1.5. Thus the peak flow can be carried by the river constriction.